The Superconducting Magnet System for the ATLAS Detector at CERN

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***Abstract* – The superconducting magnet system for the ATLAS Detector for the Large Hadron Collider at CERN is under construction since 1997. The magnet system is a hybrid of 4 superconducting magnets, a 2.5 m bore, 5.3 m long 2 T central solenoid servicing the inner detector and three 4 T toroids with global dimensions of 20m diameter, 25m long providing the magnetic field of about 1 T for the muon detectors. The operating currents of the solenoid and toroids are 8 and 20 kA respectively with a total stored energy of 1.6 GJ. The realization of the system is challenging due to its unusual size and complexity. Last year the construction of the coils and cryostats has commenced in industry. The manufacturing of the solenoid cold mass is already finished and the delivery of the individual toroid coils will commence in 2001. On-surface test of all magnets is obvious where after installation in a new cavern 100 m underground will start in 2003 leading to a commissioning of the entire magnet system in early 2005. In the paper the various magnets are introduced as well as the common infrastructure and services required. The status of the project is reviewed.**

I. INTRODUCTION

ATLAS is a new High-Energy Physics Experiment being prepared for the Large Hadron Collider at CERN. It requires the construction of a detector system able to analyse proton-proton collisions at 14 TeV level. The realisation of the detector and corresponding infrastructure is organised by the ATLAS Collaboration, a joint enterprise of about 145 institutes in 34 countries world-wide including CERN.

Particle separation and momentum measurements require a specific magnetic field distribution. For ATLAS, optimum physics performance is expected when using a combination of an axial field in the centre for the central detector region surrounded by a tangential field servicing the muon spectroscopy. Such a field configuration requires a hybrid magnet system of a solenoid making the axial magnetic field and a toroidal magnet generating the tangential magnetic field.

In order to have access to the detector interior for maintenance the toroidal magnet is split into 3 by which the solenoid and the end-cap toroids are positioned inline and within the bore of the barrel toroid. Consequently the ATLAS Magnet System [1] consists of 4 superconducting magnets:

- a Central Solenoid (CS),
- a Barrel Toroid (BT),
- two End Cap Toroids (ECT).

The magnets in the detector are shown in Figure 1. Each of the three toroids consists of 8 coils (flat pancake type) assembled radially and symmetrically around the beam axis.

The End Cap Toroid coil system is rotated by 22.5^0 with respect to the Barrel Toroid coil system to allow for radial overlap and to optimise the bending power in the interface regions of both coil systems. Obviously the solenoid is magnetically de-coupled from the toroids.

The weight of the coils and the Lorentz forces require a mechanical structure enclosing the windings and a supporting structure between the coils. As the coils are superconducting, cooling circuits and cryostats for optimum thermal insulation of the coils are a necessity. The overall dimensions of the Magnet System are determined by the Barrel Toroid that extends over a length of 26 m and an outer diameter of 20 m.

The magnet system will be installed 100 m underground in the 50,000 m^3 ATLAS cavern at collision point 1 in the LHC starting in year 2003.

A particular challenge of the ATLAS Magnet System is its size and the mixed configuration of solenoidal and toroidal coil systems and accommodating the physics requirements of a light and open structure in order to keep particle scattering at a minimum. The cryostats are dressed with detector elements and carry their weight of some 400 tons.

The performance of the toroids in terms of bending power for muon detection is characterised by the field integral $\int Bdl$ where B is the azimuthal field component and dl is a trajectory between the inner and outer radii of the toroids. The BT provides 2-6 Tm while the ECT contributes with 4-8 Tm.

The ATLAS Magnet System is large compared to other superconducting spectrometer magnets or for example the LCT and ITER type of fusion magnets which are a factor 2 to 3 smaller in size. The magnets have to be engineered and constructed in a rather short period since already in spring 2003 the installation underground must start.

For the engineering and construction CERN/ATLAS works in close co-operation with 5 magnet laboratories. CEA-Saclay (F) and INFN-LASA (I) are responsible for the engineering and industrial follow-up of the Barrel Toroid, RAL (UK) takes care of the End Cap Toroids while NIKHEF (Nl) participates considerably in the realisation. KEK (J) realises the Central Solenoid. CERN's effort is focussed mainly on the overall project management and co-ordination, as well as the infrastructure for assembly and testing, the installation underground and the common cryogenics, power and control and safety systems.

The reference design laid down in the Magnet System Technical design Reports [1] was approved in September 1997. Since then the construction has started. The major contracts for the manufacturing of the cold masses and cryostats were placed in 1998-99.

The main parameters of the magnets are listed in Table 1.

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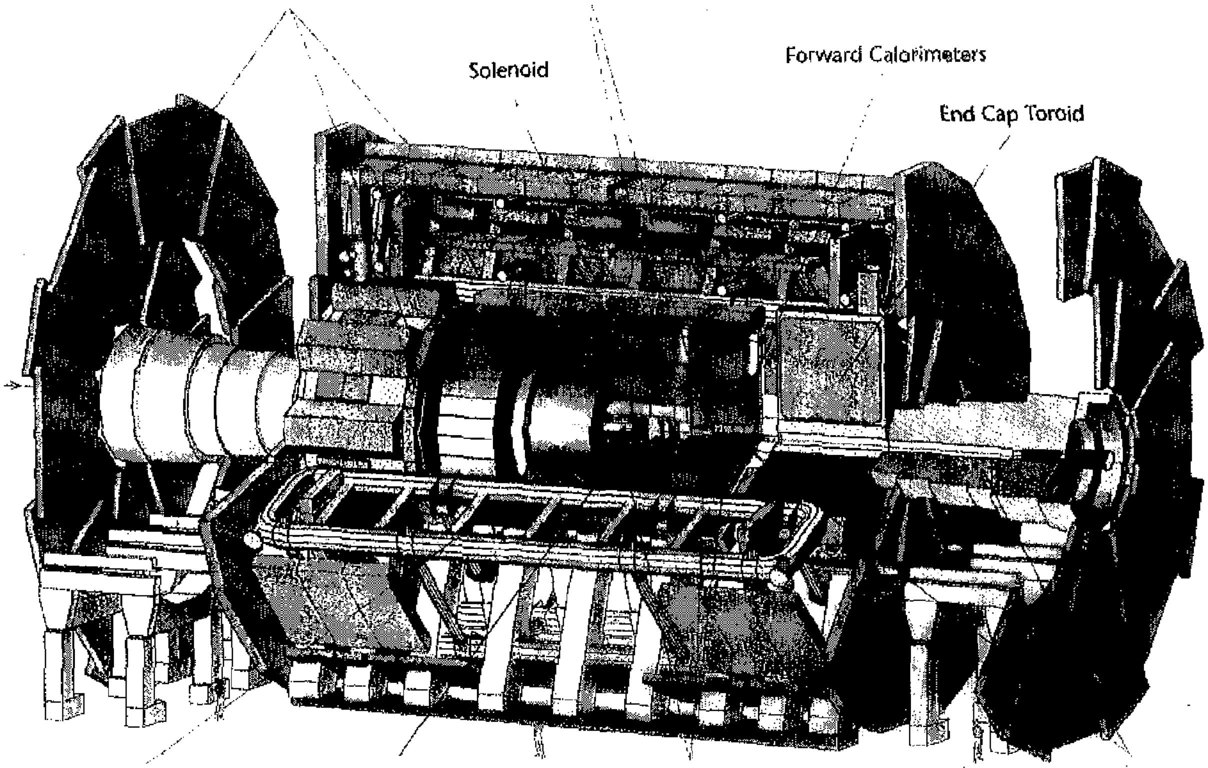

Fig. 1. Layout view of the ATLAS Detector showing the 4 superconducting magnets, BT, 2 x ECT and CS in the Magnet System. Overall dimensions are 26 m long from the left side of the left End Cap Toroid to the right side of the right End Cap Toroid and 20 m diameter for the Barrel Toroid. The Barrel Toroid encloses the two End-Cap Toroids and Central Solenoid that is positioned in the centre in the Liquid Argon Calorimeter.

II SUPERCONDUCTORS

The superconductor used is typical for detector type of magnets i.e. a composite of a flat NbTi/Cu Rutherford cable co-extruded (or plated) with aluminium with rectangular cross section. The critical current of the cable and the cross-sectional area of the Al providing stability and protection, are matched to the peak field in the coils and the stored energy respectively. In the case of the BT and ECT the cables are about 26 mm wide, a factor 2 larger than the usual 10-16 mm wide cables commonly used in accelerator dipole magnets.

For the BT and ECT conductors, the stabiliser is high purity aluminium while in the case of the CS conductor special cold-worked and doped aluminium is applied that provides increased an mechanical strength up to about 120 MPa.

The cross-section of the BT conductor as specified in Table I is shown in Figure 2. The production technology of co-extruding or plating Aluminium and NbTi/Cu cable of such large dimensions has been developed in industry during the last 4 years.

The production of the solenoid conductor was completed by Furukawa (6 km) and Hitachi (6 km) and accepted for coil winding in Dec 1998.

The BT and CT conductors are produced by two companies, Vacuumschmelze (D) and LMI (I)-ACS (CH), each producing 50 %. Meanwhile 3 units of BT conductor of 1730 m and 6 units of ECT conductor of 800 m were delivered by VAC while LMI-ACS produced 2 units of 1700m for the B0 model coil. Completion of the in total some 81 km BT/ECT conductors is foreseen for July 2000 [5].

Crucial for acceptance of produced conductor units is the critical current and inter-metallic bonding between Cu in the strands and the aluminium. At nominal field the toroid conductors operate at 30 % of the critical current with a temperature margin of 1.9 K while the solenoid operates at 20 % of Ic with a 2.7 K temperature margin. The Cu-Al bonding is required for heat and current transfer in order to guarantee proper cooling, stability and protection against conductor burn-out.

TABLE I MAIN DESIGN PARAMETERS OF THE MAGNETS

Property	Unit	Barrel Toroid	End Cap Toroids	Central Solenoid
Overall dimensions:				
Inner diameter	m	9.4	1.65	2.46
Outer diameter	m	20.1	10.7	2.63
Axial Length	m	25.3	5	5.3
Number of Coils	--	8	2 x 8	1
Mass:				
Conductor	Tons	118	2 x 20.5	3.8
Cold mass	Tons	370	2 x 160	5.4
Total assembly	Tons	830	2 x 239	5.7
Coils:				
Turns /coil	--	120	116	1173
Nominal current	KA	20.5	20.0	7.6
Magnet stored energy	GJ	1.08	2 x 0.25	0.04
Peak Field	T	3.9	4.1	2.6
Conductor:				
Overall size	Mm^2	57 x 12	41 x 12	30 x 4.25
Ratio Al:Cu:NbTi	--	28:1.3:1	19:1.3:1	15.6:0.9:1
No of strands	--	38	40	12
Strand diameter	mm	1.3	1.3	1.22
Critical current @5T,4.2K	kA	58	60	20.4
RRR Al	--	> 800	> 800	> 400
Working point at 4.5K	%	30	30	20
Temperature margin	K	1.9	1.9	2.7
No. units x length	# x m	32 x 1730	32 x 800	1 x 9100
Total length	km	56	2 x 13	10
Cooling requirements				
At 4.5 K	W	990	330	130
At 60-80 K	kW	7.4	1.7	0.50
Liquid He mass flow	g/s	410	280	6-20

III BARREL TOROID

The Barrel Toroid [2-4], see Figure 3, is build up from 8 flat racetrack coils each of which consists of two double pancake windings housed in a common aluminium alloy casing that guarantees cooling, rigidity and transfer of the magnetic forces.

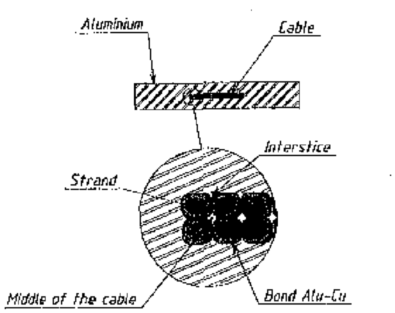

Fig. 2. Cross-section of the Aluminium stabilised NbTi/Cu superconductor for the Barrel Toroid, 57 x 12 mm^2 and 60 kA Ic at 4.2 K and 5 T.

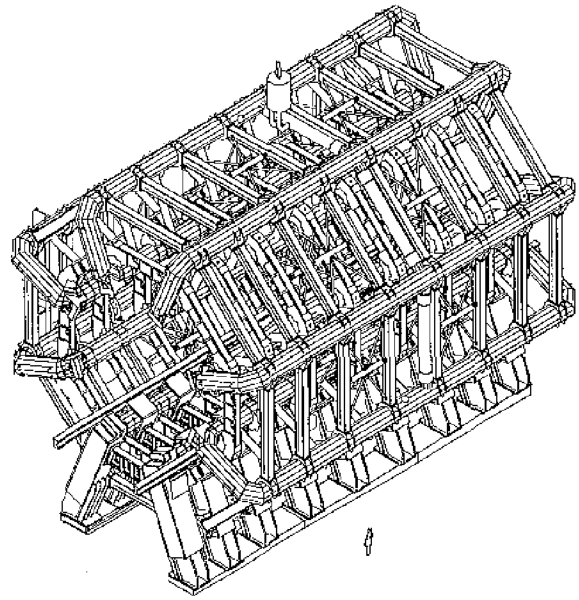

Fig. 3. Barrel Toroid magnet with 8 coils supported by a distributed and open warm structure and interconnected by a cryogenic-ring for services routing.

Double pancakes are used in both the BT and ECT. As an example, the windings and section of the cryostat of the Barrel Toroid are shown in Figure 4. Also the individual turns can be recognised. Every coil has its own cryostat that transfers the forces between the coils. The eight cryostats together with the 8 supporting rings form the warm structure guaranteeing mechanical stability in normal operation as well as under seismic loads of 0.15g.

The superconductors are kept at below 4.8 K by two-phase helium circulating in Al tubes welded on the coil casings. All electrical and cryogenic connections are routed trough a cryogenic ring linking the coils. The coil cooling circuits are in parallel. Flow meters and valves ensures the nominal flow

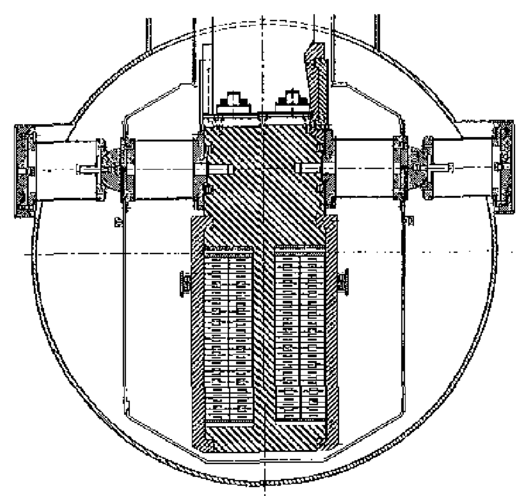

Fig. 4. Cross section of the double pancake windings of the Barrel Toroid coils and its positioning in the coil casing and the cryostat.

He required for each coil. Cold pumps located in the common feed box drive the helium through the BT and ECT coils. The thermal shields are cooled by high-pressure helium gas delivered by the refrigerator at 40 K.

In order to master the scaling up of the coils and to practice the manufacturing, a 9 m long short version of the 25 m long BT coils called Bo, see Figure 5, is presently under construction. The superconductor was made by LMI (I) using the co-extrusion facility at ACS (CH). The coil winding is finished by ANSALDO (I) and already one double pancakes was impregnated. The coil casing is in production at NFM (F). The vacuum vessel and radiation shield are produced by ZANON (I). The coils will thereafter be integrated at CEA-Saclay (F) and delivered to the test station at CERN in June 2000. The model coil is fully instrumented and will be tested under severe conditions to verify the toroids coil design.

In parallel the production of the Barrel Toroid has started and contracts in industry were placed for coil winding (ANSALDO, I), coil casing (ABB, Ch) and vacuum vessels (Felguera CM, E). Coil winding for the first 25 m long double pancakes will start in Nov 99.

The cryostat integration will be done at CERN where after the coils are tested in the test facility at CERN[12,14].

IV End-Cap Toroids

The two End-Cap Toroids [6, 7, 10] (see Figure 6) are positioned inside the Barrel Toroid at each end of the Central Solenoid. Like the Barrel Toroid, an End-Cap Toroid has 8 racetrack coils with 16 double pancake windings in a coil casing built up from Al 5083 plates. However, in contrast to the Barrel Toroid, a single cold mass is assembled from the 8 coil casings using a web structure linking the 8 coil modules and positioned in an Al 5083 vacuum vessel.

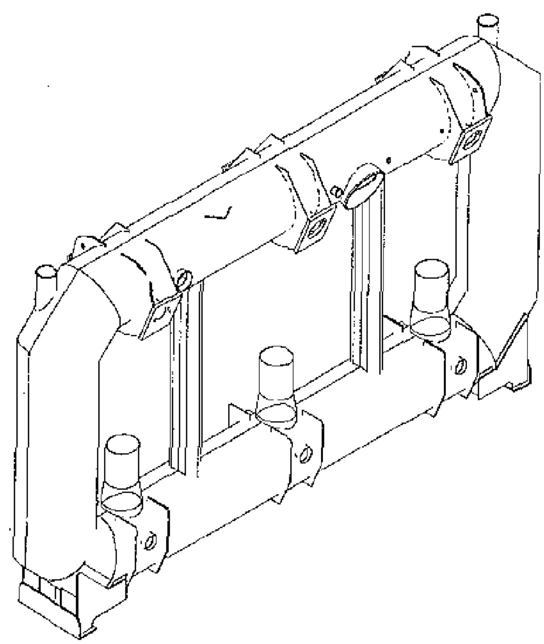

Figure 5. The 9m long, 5m wide Bo model coil currently under construction.

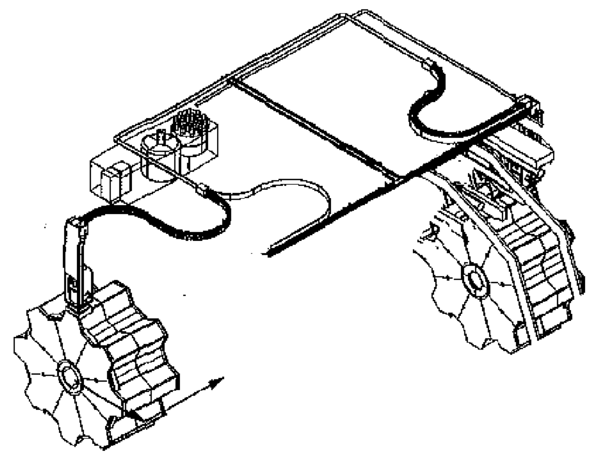

Fig. 6. System of two End Cap Toroids. In each Toroid of 10.7 m dia, 5 m axial length, there is one cold mass of 8 coils in a common Al cryostat visible through the castlellated outer shell. Services are connected via the turrets.

The magnetic overlap at the ECT-BT interface causes a Lorentz force pulling the End Cap Toroid from both sides into the BT bore. The axial forces are transferred to the BT vacuum vessels via 8 transfer points at each side. This causes a buckling stress in the BT vacuum vessels. The current and He cooling are provided from the top and connections are made in a turret. While the Barrel Toroid is fixed on a double row of stiff feet, the End Cap Toroids are movable on a rail system facilitating the movement and parking of the ECT magnets for the case access is required to the detector centre.

In early 1999 contracts were placed with HMA Power Systems (Nl) and Royal Schelde Exotech (Nl) for the manufacturing of the 2 cold masses and vacuum vessels respectively. First deliveries of cold masses and vacuum vessels are expected in 2001. Other parts of the system i.e the radiation shield, superinsulation and coil supports, will be procured in year 2000. The integration work at CERN will commence in 2001. In spring 2003 the two ECT's should be ready for testing on the surface and their installation is scheduled for late 2004.

V Central Solenoid

The Central Solenoid [8,9] with a length of 5.3 m and a bore of 2.5 m, see Figure 7, provides 2 T axial magnetic field. In order to minimise the material thickness, the solenoid is mounted in the vacuum vessel of the Liquid Argon Calorimeter and is only 45 mm thick. The 12 mm thick outer support cylinder is only feasible when the conductor itself is strong enough to take a big part of the hoop stress. The technology for conductor strengthening has been successfully developed by Furukawa and Hitachi in collaboration with KEK. Doping of the pure Al leads to acceptable stress levels of about 120 MPa while keeping a relatively high RRR. The coil is indirectly cooled using forced flow two-phase helium

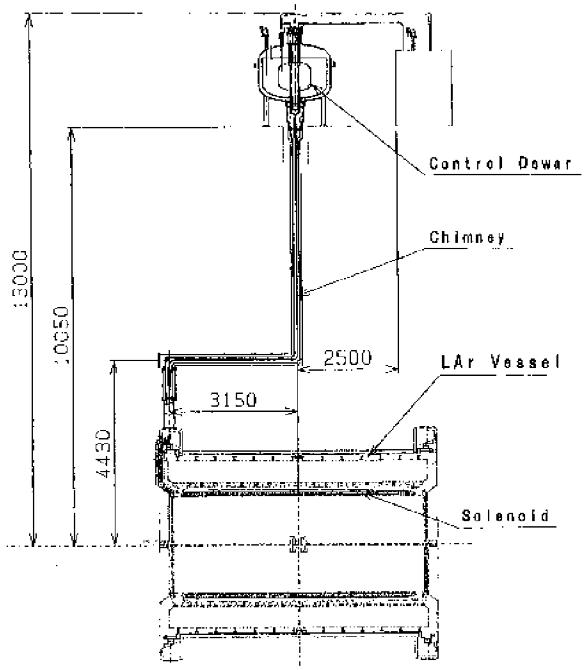

Fig. 7. Central Solenoid, 2.5 m bore, 5.3 m length, mounted in a common cryostat with the Liquid Argon calorimeter.

in tubes welded on the outer support cylinder. The superconducting bus bars and cryo-lines connect the coil through a 10-m long chimney to the proximity cryogenics positioned on top of the detector.

The solenoid project is well underway. The cold mass has been produced by Toshiba (J) using conductors and support cylinder manufactured by Hitachi (J), Furukawa (J) and Oxford Instruments (UK) respectively. Next are the delivery of the cryogenics and the test of the coil in Japan by Toshiba (J). Thereafter the coil will be transported to CERN for integration with the Liquid -Argon cryostat.

VI Cryogenics

The 4.8 K operating temperature in the toroid coils is achieved by circulating He by an array of pumps providing up to 1200 g/s through tubes welded on the cold masses. The Central Solenoid is directly coupled via a dewar to the refrigerator where as the three Toroids are fed from a Helium pump and distribution unit, see Figure 8. This common proximity cryogenics system for the toroids is now under detailed design at RAL leading to a specification by summer 2000 where after the units will be procured. Installation in the underground cavern must commence by Oct 2002.

The unit is supplied with He from two central refrigerators providing 6 kW at 4.5 K and some 15kW at 80K respectively. [9,11]. This covers the thermal budget of the four magnets i.e. 2.5 kW isothermal refrigeration at 4.5 K, 10.5 g/s Helium flow for the current leads, and some 14 kW for the 40-80 K thermal shields. In addition, a dedicated pre-cooling unit with LN_2/He heat exchanger using 24 m^3 LN_2 per day is installed.

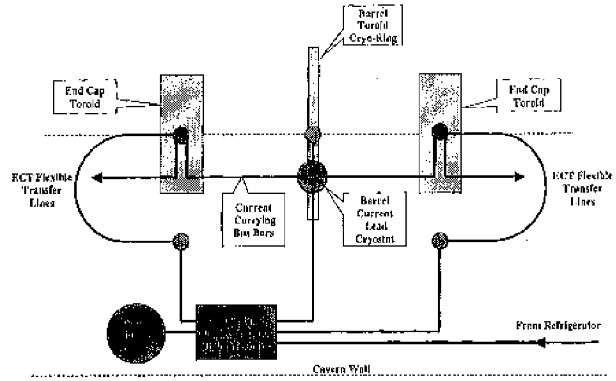

Fig. 8. Layout of the proximity cryogenics system feeding the Toroids. Dewar and the valves-pumps-distribution-separation cryostats are located at the wall of the cavern from which three cryogenic lines feed the three Toroids.

The refrigerator cold box, LN_2 pre-cooling unit and auxiliary equipment are housed underground in the side cavern while the He screw compressors, purification and recuperation systems and the 15000 m^3 He and 100 m^3 LN_2 storage tanks are installed at the surface.

VII Power and Protection

The Toroids, in fact all the 32 coils, can be electrically connected in series, see Figure 9. In order to safely handle the stored energy during quench, voltage limiting run-down resistors are connected across the Barrel and End Cap Toroids.

The Central Solenoid has a separate 8 kA supply due to its different current rating. The toroids circuit consist of the power supply, a circuit breaker, some 300 m bus bars, He cooled current leads and the diode-resistor combination connected across the supply/breaker and the coils. The ramp-up time of the Toroids is 2-3 hours.

The Toroids have an active quench protection system. After quench detection, heaters in the coils are fired in order to safely dissipate the stored energy in the coil windings without overheating the conductors. The effect of a quench in the worst failure mode (no detection, no heaters, no switch opening and power supply on) was analysed carefully [15].

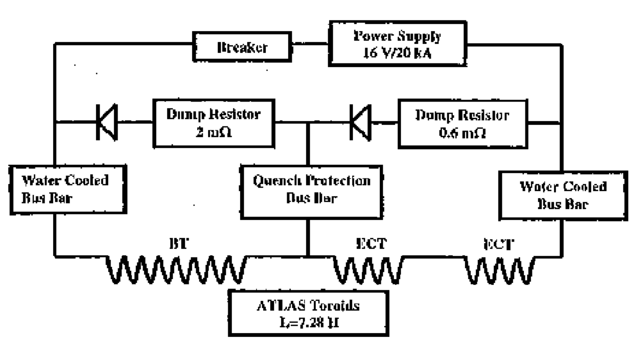

Fig. 9. Electrical circuit of the Barrel and End-Cap Toroids magnet system.

A peak coil temperature of 300 K may be expected while during a normal quench the highest temperature remains below 130 K. The power supplies, breakers, diodes and run-down resistor are located in the side cavern.

The magnet control system is interfaced with the top level ATLAS control room, the local refrigeration plant, the central cryogenics control room, the LHC control room, the detector and experiment safety system. The safety of the magnets is ensured by independent hard-wired logic and PLC circuits controlling slow dump (2 hours ramp-down) or fast dump (<100 s) of the stored energies. Safety sensors and channels are doubled and UPS units will be installed.

VIII Installation and Testing

For obvious reasons all coils will be tested on surface before installation underground. The thorough performance and acceptance tests will be performed at CERN in a test facility especially build up for this purpose [14, 16].

After installation in the cavern, see Figure 10, the Barrel Toroid will be tested and then integrated with both End Cap Toroid Magnets and the Central Solenoid.

IX Final Remarks

The ATLAS Collaboration is producing since 2 years the superconducting magnet system for its detector. The magnet system comprising a solenoid and 3 toroids is in size the largest integrated superconducting magnet known today but has to be constructed in a rather short period of 7 years. The manufacturing is well underway. Contracts for the cold masses and cryostats have been placed, the solenoid cold mass is already complete and the delivery of the toroid cold masses and cryostats will start in year 2000. In the design the focus is now on the cryogenics, electrical and control systems as well as all the assembly and integration work to be done in the coming years.

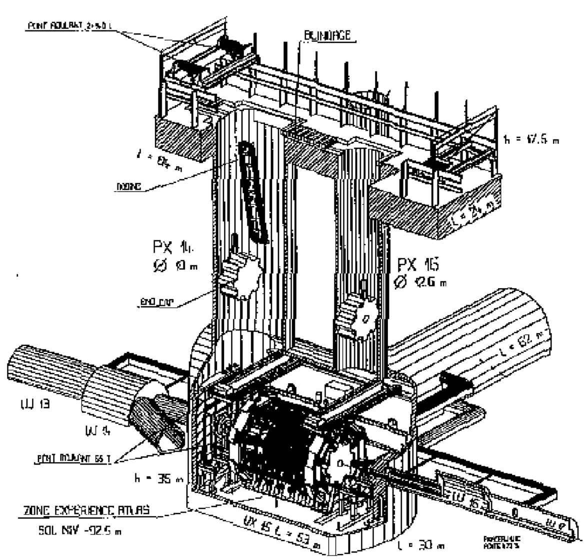

Figure 10. The ATLAS site at point 1 in the LHC. The underground caverns and the shafts are shown. It is illustrated how the 8 Barrel Toroid coils and the two assembled End-Cap Toroids will be lowered into the cavern.

Today about 60 % of the 140 MCHF budget of the magnet system is committed. The major contracts and agreements for in-kind deliveries were agreed within the financial envelopes. Also the schedule of the Solenoid and End-Cap Toroids is satisfactory. The hardware is produced in time and with about 1 year of float before it starts to become critical. For the Barrel Toroid the schedule is very tight and shows no contingency. Nevertheless, the present schedule is still compatible with a start of the underground installation in March 2003 and commissioning of the system in early 2005.

Acknowledgement

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